

Year 12 3 UNIT
Term 2 – Assessment - 1999

INSTRUCTIONS:

- Time allowed: 85 minutes
- This is an open book test
- Show all necessary working

QUESTION 1 (START A NEW PAGE)

(a) A container of soup is removed from the top of a campfire. The soup is left to stand for several minutes in surrounds of a constant 15° . The soup cools at a rate proportional to the difference between its temperature (T) and the surrounding

temperature (S), i.e. $\frac{dT}{dt} = -0.05(T - S)$.

(i) Show that $T = S + Ae^{-0.05t}$ is a solution of $\frac{dT}{dt} = -0.05(T - S)$.

(ii) If the soup is initially at 85° , write down the values of S and A .

(iii) Find the temperature of the soup after 10 minutes. (Give your answer to the nearest degree)

(b) All the letters of the word EQUATION are arranged at random in a line. How many arrangements are possible if

- (i) there are no restrictions
- (ii) there is a consonant at each end.
- (iii) all the vowels are together.

QUESTION 2 (START A NEW PAGE)

(a) The velocity ($v \text{ ms}^{-1}$) of an object at position x metres is given by $v = 2\sqrt{x(x-20)}$.

- (i) Find the values of x for which motion is possible.
- (ii) Show that the acceleration is given by $\ddot{x} = 4(x-10)$.
- (iii) Find the velocity when the acceleration is 80 ms^{-2} .

(b) The position of an object at time t is given by $x = \sin^2 6t$. Prove that the acceleration $\ddot{x}$ can be expressed in the form $\ddot{x} = -n^2(x-b)$.

QUESTION 3 (START A NEW PAGE)

(a) A toy rocket is fired from the top of a 100 metre high cliff and its position at time t is given by the equations $x = 30t$ and $y = -5t^2 + 40t + 100$. Find the

- (i) greatest height above the ground that the rocket reaches.
- (ii) distance from the base of the cliff to the point where the rocket hits the ground.

(b) Three notes are chosen at random from an envelope containing three \$5, five \$10 and two \$20 notes. Find the number of different selections that can be made if

- (i) they are all different denominations
- (ii) their total value is \$30

QUESTION 4 (START A NEW PAGE)

(a) 100 white and 100 black marbles are mixed together. Some are placed in urn A while the rest are placed in urn B. You are told that the probability of selecting a white marble from urn A is $\frac{2}{3}$ and if a white marble is placed in urn B then the probability of selecting a black marble from urn B is also $\frac{2}{3}$. Find the number of each colour originally in urn A.

(b) An object moving with SHM has its position at time t given by $x = a \cos(nt + \alpha)$. The object is initially 6 metres to the right of the origin. If the period of its motion is 8 minutes and its maximum speed is $3\pi \text{ m/min}$, find

- (i) the values of n , a and α .
- (ii) the first time when the object passes through the origin.

QUESTION 5 (START A NEW PAGE)

(a) Four girls (Anne, Betty, Carole and Dianne) and four boys (Peter, Quentin, Ross and Stuart) are seated around a circular table.

(i) Find the number of different random arrangements that are possible

Find the probability that in a random arrangement

(ii) all the girls are seated together.

(iii) the boys and girls sit in alternate seats.

(b) In a colony of 1 million birds, the number infected with a disease after time t weeks is given by $N = \frac{1000000}{100 + 1900e^{-0.05t}}$.

(i) How many birds are initially infected?

(ii) How many birds will ultimately be infected?

(iii) Show that $\frac{dN}{dt} = \frac{N(10000 - N)}{200000}$.

(iv) How many birds will be present when the rate of infection is greatest?

infected

QUESTION 6 (START A NEW PAGE)

(a) Show that $\frac{d}{dx} \left[\cos^{-1} \left(\frac{x^2}{16} \right) \right] = \frac{-2x}{\sqrt{256 - x^4}}$.

(b) A particle initially at rest at $x = 4$, has an acceleration towards the origin given by

$$\ddot{x} = -4 \left(x + \frac{256}{x^3} \right) \text{ for } x > 0.$$

(i) Show that the velocity v is given by $v^2 = 4 \left(\frac{256}{x^2} - x^2 \right)$.

(ii) Explain why the velocity is negative for $0 < x < 4$.

(iii) Using part (a) find an expression for the time taken to reach position x .

(iv) Find the time for the particle to reach $x = 2\sqrt{2}$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

THIS IS THE END

QUESTION 1

(a)(i) $\frac{dT}{dt} = -0.05 A e^{-0.05t}$
 $\frac{dT}{dt} = -0.05(T-5)$ since $A e^{-0.05t} = T-5$

(ii) $S = 15$

$t = 0, T = 85$

$\therefore 85 = 15 + A e^0$

$A = 70$

$\therefore A = 70, S = 15$

(iii) $T = 15 + 70 e^{-0.05t}$

$t = 10, T = 15 + 70 e^{-0.5}$

$= 57.46^\circ$

$= 57^\circ$ (to nearest degree)

(i) No arrangements = $8!$

(ii) Arrange consonant = 3.2

arrange int = $6!$

Total = $3.2.6!$

(iii) arrange vowels = $5!$

arrange vowel group + 3 others = $4!$

Total = $5!.4!$

QUESTION 2

(a)(i) $x(x-20) \geq 0$

$\therefore x \leq 0, x \geq 20$

(ii) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$\frac{1}{2} v^2 = 2x(x-20)$

$= \frac{d}{dx} (2x^2 - 40x)$

$= 4x - 40$

$\ddot{x} = 4(x-10)$

(14) $x = 80 \Rightarrow 80 = 4(x-10)$

$20 = x - 10$

$x = 30$

$\therefore v = 2\sqrt{30.10}$

$= 20\sqrt{3} \text{ m/s}$

(b) $x = \sin^2 6t$

$\dot{x} = 2 \sin 6t \cdot 6 \cos 6t$

$= 12 \sin 6t \cos 6t$

$= 6 \sin 12t$

$\ddot{x} = 72 \cos 12t$

$= 72(1 - 2 \sin^2 6t)$

$= 72(1 - 2x)$

$= -144(x - \frac{1}{2})$

or

$\dot{x} = 12 \sin 6t \cos 6t$

$\ddot{x} = 12 \{ (\sin 6t)(-6 \sin 6t) + (6 \cos 6t)(\cos 6t) \}$

$= 12 \{ -6 \sin^2 6t + 6 \cos^2 6t \}$

$= 72 \{ -\sin^2 6t + 1 - \sin^2 6t \}$

$= 72(1 - 2 \sin^2 6t)$

$= 72(1 - 2x)$

$= -144(x - \frac{1}{2})$

QUESTION 3.

(a)(i) $y = -10t + 40$

$\dot{y} = 0 \Rightarrow 10t = 40$

$t = 4 \text{ s.}$

$\therefore y = -5(4)^2 + 40(4) + 100$

$= 180 \text{ m.}$

(ii) $y = 0 \Rightarrow -5t^2 + 40t + 100 = 0$

$t^2 - 8t - 20 = 0$

$(t-10)(t+2) = 0$

$t = 10 \quad (t > 0)$

$\therefore x = 30(10)$

$= 300 \text{ m.}$

(b) $\{ 5, 5, 5, 10, 10, 10, 10, 10, 20, 20 \}$

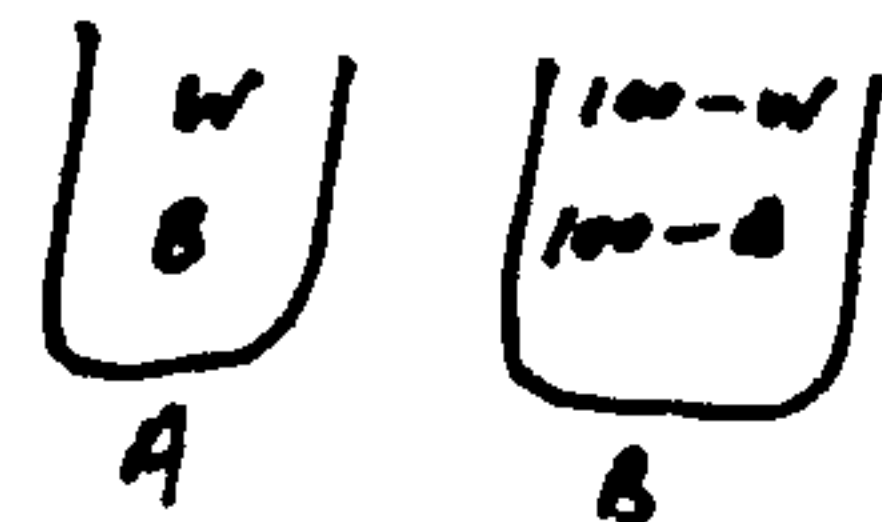
(i) All diff = ${}^3C_1 \cdot {}^5C_1 \cdot {}^2C_1$

$= 30$

Q3(b)(ii) $(\$10, \$10, \$10)$ or $(\$5, \$5, \$20)$
 $N = x_1 + x_2 + x_3$
 $= 16$

QUESTION 4.

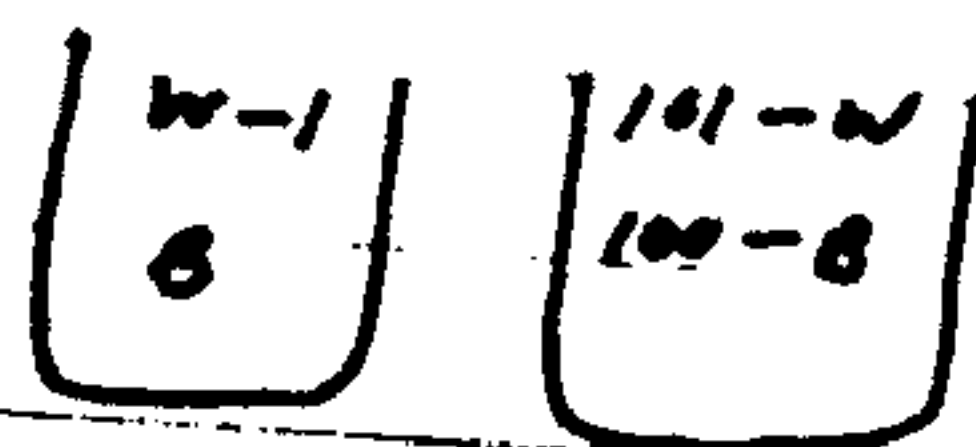
(a) In urn A let W = number of white & B = number of black



$$\frac{W}{W+B} = \frac{2}{3}$$

$$3W = 2W + 2B$$

$$W = 2B \quad \text{--- (1)}$$



$$\frac{100-B}{201-W-B} = \frac{2}{3}$$

$$300 - 3B = 402 - 2W - 2B$$

$$2W - B = 102 \quad \text{--- (2)}$$

Sub. (1) into (2)

$$4B - B = 102$$

$$3B = 102$$

$$B = 34$$

$$+ W = 68$$

$\therefore 68$ white + 34 black.

(b) (i) period = $2\pi/\omega$

$$= 2\pi/\omega = 8$$

$$\omega = \pi/4$$

$$\text{max speed} = \omega a$$

$$3\pi = \frac{\pi}{4} a$$

$$a = 12$$

$$t=0, x=6$$

$$6 = 12 \cos \alpha$$

$$\cos \alpha = \frac{1}{2}$$

$$\alpha = \pi/3$$

$$x = a \cos\left(\frac{\pi}{4}t + \alpha\right)$$

$$\text{or } \dot{x} = -\frac{\pi a}{4} \sin\left(\frac{\pi}{4}t + \alpha\right)$$

$$\therefore 3\pi = -\frac{\pi a}{4}$$

$$a = -12$$

$$t=0$$

$$+6 = -12 \cos \alpha$$

$$\cos \alpha = -\frac{1}{2}$$

$$\alpha = 2\pi/3$$

Q4(b)(i) $x = 12 \cos\left(\frac{\pi}{4}t + \frac{\pi}{3}\right)$

$$0 = 12 \cos\left(\frac{\pi}{4}t + \frac{\pi}{3}\right)$$

$$\frac{\pi}{4}t + \frac{\pi}{3} = \frac{\pi}{2}$$

$$\frac{\pi}{4}t = \frac{\pi}{6}$$

$$t = \frac{2}{3} \text{ min}$$

$x = -12 \cos\left(\frac{\pi}{4}t + \frac{2\pi}{3}\right)$

$$0 = -12 \cos\left(\frac{\pi}{4}t + \frac{2\pi}{3}\right)$$

$$\frac{\pi}{4}t + \frac{2\pi}{3} = \frac{3\pi}{2} \quad | \times 12$$

$$\frac{\pi}{4}t = \frac{5\pi}{2} - \frac{2\pi}{3}$$

$$= \frac{5\pi}{6}$$

$$t = \frac{5\pi}{6} \cdot \frac{4}{\pi}$$

$$= 3\frac{1}{3} \text{ min}$$

QUESTION 5

(a) (i) N arrangements = $7!$

(ii) arrange girls = $4!$

arrange girl group + 4 boys = $4!$

$$\text{Total} = 4!4!$$

$$\therefore \text{Prob} = \frac{4!4!}{7!}$$

(iii) place girls = $3!$

place boys = $4!$

$$\text{Total} = 3!4!$$

$$\text{Prob} = \frac{3!4!}{7!}$$

(b) (i) $t=0, N = \frac{10^6}{100+1900}$
 $= 500$

(ii) $t \rightarrow \infty, e^{-0.05t} \rightarrow 0$
 $\therefore N \rightarrow \frac{10^6}{100}$

$$\therefore N = 10000$$

$$\begin{aligned}
 \text{Q5 (iii)} \quad N &= 10^6 (100 + 1900e^{-0.05t})^{-1} \\
 \frac{dN}{dt} &= 10^6 \cdot -1 (100 + 1900e^{-0.05t})^{-2} \cdot -0.05 \cdot 1900e^{-0.05t} \\
 &= \frac{10^6 \times 0.05 \times 1900 e^{-0.05t}}{(100 + 1900e^{-0.05t})^2} \\
 &= \frac{95 \times 10^6 e^{-0.05t}}{(100 + 1900e^{-0.05t})^2}
 \end{aligned}$$

Now

$$\frac{N}{10^6} = \frac{1}{100 + 1900e^{-0.05t}} \quad \text{--- (1)}$$

$$100 + 1900e^{-0.05t} = \frac{10^6}{N}$$

$$1900e^{-0.05t} = \frac{10^6}{N} - 100$$

$$e^{-0.05t} = \frac{10^6 - 10^2 N}{1900 N} \quad \text{--- (2)}$$

$$\begin{aligned}
 \therefore \frac{dN}{dt} &= 95 \times 10^6 \left(\frac{10^6 - N}{19 N} \right) \cdot \left(\frac{N}{10^6} \right)^2 \\
 &= 95 \times 10^6 \left(\frac{10000 - N}{19 N} \right) \cdot \frac{N^2}{10^{12}} \\
 &= \frac{95 (10000 - N) N}{19 \times 10^6} \\
 &= \frac{N (10000 - N)}{200000}
 \end{aligned}$$

(iv) rate greatest when $N = 5000$

QUESTION 6

(a) let $u = x^2/16$

$$\therefore \frac{d}{dx} \left(\cos^{-1} \frac{x^2}{16} \right) = \frac{d}{du} \left(\cos^{-1} u \right) \cdot \frac{du}{dx}$$

$$= \frac{-1}{\sqrt{1-u^2}} \cdot \frac{2x}{16}$$

$$= \frac{-x}{8\sqrt{1-\frac{x^4}{256}}}$$

$$\text{Q6(b)(iii)} \quad \frac{dx}{dt} = \frac{-2\sqrt{256-x^4}}{x}$$

$$\frac{dt}{dx} = \frac{-x}{2\sqrt{256-x^4}}$$

$$t = \int \frac{-x}{2\sqrt{256-x^4}} dx$$

$$= \frac{1}{4} \int \frac{-2x}{\sqrt{256-x^4}} dx$$

$$t = \frac{1}{4} \cos^{-1} \left(\frac{x^2}{16} \right) + C$$

$$t=0, x=4 \Rightarrow 0 = \frac{1}{4} \cos^{-1}(1) + C$$

$$\therefore C = 0$$

$$\therefore t = \frac{1}{4} \cos^{-1} \left(\frac{x^2}{16} \right)$$

$$\begin{aligned}
 \text{(iv)} \quad x &= 2\sqrt{2}, \quad t = \frac{1}{4} \cos^{-1} \left(\frac{8}{16} \right) \\
 &= \frac{1}{4} \cos^{-1} \left(\frac{1}{2} \right) \\
 &= \frac{1}{4} \cdot \frac{\pi}{3}
 \end{aligned}$$

$$t = \pi/12$$

$\neq$